

DWG No. 9496337

SH 1

APPLICATION		REVISIONS			
NEXT ASSY	USED ON	REV	DESCRIPTION	DATE	APPROVAL
9496336	9496336				

SEE TABLE 1
INTERPRET CCs IAW DOD-STD-2101
SUMMATION OF CCs
CRITICAL <u>NONE</u> MAJOR <u>10</u>
CLASSIFICATION OF CHARACTERISTICS

CONTRACT NO. _____		DISTRIBUTION STATEMENT A. APPROVED FOR PUBLIC RELEASE: DISTRIBUTION UNLIMITED																							
CONTRACTOR _____																									
APPROVED _____		DATE _____		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
DRAWN _____		REVISION STATUS OF SHEETS																							
UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES GENERAL TOLERANCES		NAVAL SURFACE WARFARE CENTER INDIAN HEAD DIVISION INDIAN HEAD, MD 20640-5035										U. S. AIR FORCE													
XXX DECIMALS ± .005												SWIVEL, ASSEMBLY													
XX DECIMALS ± ---		4330 MURPHY 95-11-06																							
ANGLES ± ---		DRAWN MIJAN 95-11-06																							
INTERPRET DRAWING IAW DOD-STD-100		APPROVED FOR NAVSEA 19 <i>R. Duran</i> 4 April 97										SIZE A		CAGE CODE 98747		DWG No. 9496337									
DO NOT SCALE THIS DRAWING												SCALE:				SHEET 1 OF 12									

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SEL LAY 1-244

1. SCOPE

1.1 THIS DRAWING ESTABLISHES THE REQUIREMENTS FOR THE MANUFACTURE AND OF A SWIVEL ASSEMBLY.

2. APPLICABLE DOCUMENTS

QQ-W-428 MIL-~~W~~-18375 MIL-I-23053
MIL-C-83488 MIL-STD-129 DOD-STD-2101
MS20664 MS51844

3. REQUIREMENTS

3.1 GENERAL

3.1.1 CONSTRUCTION: MATERIAL, PARTS AND ASSEMBLIES SHALL BE MANUFACTURED IAW THIS DRAWING AND ALL DRAWINGS AND SPECIFICATIONS LISTED HEREON.

3.1.2 FIRST ARTICLE: UNLESS OTHERWISE SPECIFIED IN THE CONTRACT, THE CONTRACTOR IS RESPONSIBLE FOR THE FABRICATION AND TESTING OF A FIRST ARTICLE SAMPLE. SAMPLE SHALL BE MANUFACTURED IAW 3.1.1 USING THE FABRICATION PROCEDURES AND FACILITIES PROPOSED FOR PRODUCTION. ACCEPTANCE OF THE FIRST ARTICLE SAMPLE QUALIFIES THE FABRICATION PROCEDURES AND THE SPECIFIC FACILITIES BY WHICH MANUFACTURE AND TEST OF THE SAMPLE IS ACCOMPLISHED. AFTER MANUFACTURE AND ACCEPTANCE OF THE FIRST ARTICLE SAMPLE, NO CHANGES SHALL BE MADE IN THE FABRICATION PROCEDURES WITHOUT THE APPROVAL OF THE PROCURING AGENCY. WHEN NEW OR ALTERED FACILITIES ARE USED IN PRODUCTION, A NEW SAMPLE SHALL BE SUBMITTED AT THE DISCRETION OF THE PROCURING AGENCY FOR ACCEPTANCE AS SPECIFIED IN 4.3.

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3.2 PERFORMANCE AND PRODUCT CHARACTERISTICS

3.2.1 VISUAL AND MECHANICAL: PARTS OF THE SWIVEL ASSEMBLY SHALL ROTATE FREELY.

3.2.2 LINK COIL: THE FORMED COIL OF THE LINK SHALL HAVE A MINIMUM OF 1 1/4 COILS AS SHOWN IN FIGURE 2.

3.2.3 LOAD REQUIREMENT

(M101)

3.2.3.1 MINIMUM LOAD REQUIREMENT: THE SWIVEL ASSEMBLY OF FIGURE 3 SHALL SUSTAIN A MINIMUM LOAD OF 235.0 POUNDS WHEN TESTED IAW 4.5.1.

(M102)

3.2.3.2 MAXIMUM LOAD REQUIREMENT: THE SWIVEL ASSEMBLY MUST SEPARATE BEFORE A LOAD OF 325.0 POUNDS, WHEN TESTED IAW 4.5.1.

(M103)

3.2.4 SEPARATION MODE REQUIREMENT: THE SWIVEL ASSEMBLY MUST SEPARATE BY PULLING THE SWAGED BALL COMPLETELY THROUGH THE COILS OF THE LINK.

3.3 WORKMANSHIP: THE SWIVEL ASSEMBLY INCLUDING ALL PARTS AND ACCESSORIES, SHALL BE CONSTRUCTED AND FINISHED IN A MANNER TO ASSURE COMPLIANCE WITH ALL REQUIREMENTS OF THIS DRAWING. PARTICULAR ATTENTION SHALL BE PAID TO NEATNESS AND THOROUGHNESS OF ASSEMBLY, AND FREEDOM, OF PARTS, FROM FOREIGN MATERIALS, BURRS AND SHARP EDGES. THE WORKMANSHIP EXHIBITED IN THE ACCEPTABLE FIRST ARTICLE SAMPLES SHALL BECOME THE MINIMUM WORKMANSHIP STANDARD FOR PRODUCTION ITEMS.

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4 QUALITY ASSURANCE PROVISIONS

4.1 RESPONSIBILITY FOR INSPECTION: UNLESS OTHERWISE SPECIFIED IN THE CONTRACT, THE CONTRACTOR IS RESPONSIBLE FOR THE PERFORMANCE OF ALL INSPECTION REQUIREMENTS AS SPECIFIED HEREIN. EXCEPT AS OTHERWISE SPECIFIED IN THE CONTRACT, THE CONTRACTOR MAY USE HIS OWN OR ANY OTHER FACILITY SUITABLE FOR THE PERFORMANCE OF THE INSPECTION REQUIREMENTS SPECIFIED HEREIN.

UNLESS DISAPPROVED BY THE GOVERNMENT. THE GOVERNMENT RESERVES THE RIGHT TO PERFORM ANY OF THE INSPECTIONS SET FORTH IN THIS DOCUMENT WHERE SUCH INSPECTIONS ARE DEEMED NECESSARY TO ASSURE SUPPLIES AND SERVICES CONFORM TO PRESCRIBED REQUIREMENTS.

4.1.1 PRODUCT INSPECTION: THE MINIMUM INSPECTION REQUIREMENTS FOR THE PRODUCT ARE SPECIFIED BY THE CLASSIFICATION OF CHARACTERISTICS ON THIS DRAWING. OTHER APPLICABLE DRAWINGS AND BY THE SUPPLEMENTAL QUALITY ASSURANCE PROVISIONS OF THE CONTRACT. THIS INSPECTION DOES NOT RELIEVE THE CONTRACTOR OF THE RESPONSIBILITY FOR MEETING ALL OTHER SPECIFICATION AND DRAWING REQUIREMENTS.

4.2 CLASSIFICATION OF INSPECTIONS: THE INSPECTION REQUIREMENTS SPECIFIED HEREIN ARE CLASSIFIED AS FOLLOWS:

A. FIRST ARTICLE INSPECTION (SEE 4.3.1).

B. QUALITY CONFORMANCE INSPECTION (SEE 4.3.2).

4.3 SAMPLING AND INSPECTION

4.3.1 FIRST ARTICLE INSPECTION: PRIOR TO INITIATING REGULAR PRODUCTION, A FIRST ARTICLE SAMPLE OF EIGHTY (80) SWIVEL ASSEMBLIES SHALL BE MANUFACTURED

IAW 3.1.2. THIS SAMPLE SHALL BE INSPECTED IAW WITH TABLE 1. FAILURE OF ANY SAMPLE TO PASS THIS INSPECTION SHALL CAUSE REJECTION OF THE FIRST ARTICLE SAMPLE. FURTHER PRODUCTION OF THE ITEM BY THE CONTRACTOR PRIOR TO APPROVAL OF THE PROCURING AGENCY SHALL BE AT THE CONTRACTORS RISK.

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4.3.2 QUALITY CONFORMANCE INSPECTION

4.3.2.1 INSPECTION LOT: INSPECTION OF EACH LOT OF SWIVEL ASSEMBLIES SUBMITTED FOR ACCEPTANCE SHALL BE PERFORMED IAW TABLE 1. FOR ACCEPTANCE PURPOSES THE LOT SIZE SHALL BE LIMITED TO A MAXIMUM OF 10,000 UNITS.

4.3.2.2 SUBMISSION OF LOT: AT THE TIME A LOT IS SUBMITTED TO THE GOVERNMENT FOR ACCEPTANCE, THE CONTRACTOR SHALL SUPPLY THE FOLLOWING INFORMATION:

A. CERTIFICATION THAT THE LOT COMPLIES WITH ALL OF THE QUALITY ASSURANCE PROVISIONS SPECIFIED WITHIN THIS DOCUMENT.

B. NUMBER OF UNITS CONTAINED IN THE LOT.

C. CERTIFICATES OF COMPLIANCE FROM SUPPLIERS ON ALL MATERIALS AND PROCESSES PURCHASED BY THE CONTRACTOR WHEN SUCH MATERIALS AND PROCESSES ARE CONTROLLED BY GOVERNMENT OR COMMERCIAL SPECIFICATIONS AND STANDARDS REFERENCED IN THIS DOCUMENT. THE GOVERNMENT RESERVES THE RIGHT TO CHECK SUCH ITEMS AS DEEMED NECESSARY TO DETERMINE THE VALIDITY OF THE CERTIFICATES.

4.3.3 CLASSIFICATION OF CHARACTERISTICS (CC) ON DRAWINGS: UNLESS OTHERWISE SPECIFIED IN THE CONTRACT OR BY THE GOVERNMENT REPRESENTATIVE, CC'S ON DRAWINGS SHALL BE INSPECTED IAW DOD-STD 2101.

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4.4 INSPECTION EQUIPMENT, FACILITIES AND CONDITIONS

4.4.1 INSPECTION EQUIPMENT AND FACILITIES: THE CONTRACTOR SHALL FURNISH AND MAINTAIN ALL NECESSARY INSPECTION EQUIPMENT AND FACILITIES FOR PERFORMING THE REQUIRED INSPECTIONS.

4.4.2 INSPECTION CONDITIONS: UNLESS OTHERWISE SPECIFIED ALL INSPECTIONS SHALL BE PERFORMED AT ROOM AMBIENT CONDITIONS.

4.5 INSPECTION PROCEDURES

4.5.1 LOAD TEST: THE SWIVEL ASSEMBLY OF FIGURE 3 SHALL BE MOUNTED IN A TENSILE TESTING APPARATUS USING .375 DIA PINS TO APPLY FORCE TO THE LINK AND THE SWAGED BALL ASSEMBLY. THE SWIVEL SHALL BE TENSILE TESTED AT A CROSS HEAD SPEED OF 5 ± 1 CM/MIN. THE SWAGED BALL ASSEMBLY SHALL SEPARATE FROM THE LINK AT A LOAD OF NO LESS THAN 235.0 LB AND AT A LOAD OF NO MORE THAN 325.0 LB. THE SEPARATION MODE MUST MEET THE REQUIREMENTS OF 3.2.4. THIS TEST SHALL BE PERFORMED ON 1% (ROUNDED TO NEAREST WHOLE NUMBER) OF THE LOT OR A MINIMUM OF 10 SWIVELS, WHICHEVER IS GREATER. FAILURE OF ONE OR MORE SWIVELS TO PASS THIS TEST SHALL CAUSE REJECTION OF THE LOT. LOAD TEST SAMPLES DO NOT COUNT AS DELIVERABLE ITEMS.

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5. PREPARATION FOR DELIVERY

5.1 PACKAGING:

5.1.1 LEVEL C - THE SWIVEL ASSEMBLY SHALL BE PACKAGED IN A MANNER WHICH WILL AFFORD ADEQUATE PROTECTION FROM DETERIORATION AND PHYSICAL DAMAGE DURING SHIPMENT FROM THE CONTRACTOR OR SUPPLY SOURCE TO THE FIRST RECEIVING ACTIVITY FOR IMMEDIATE USE OR OR NEXT LEVEL OF ASSEMBLY.

5.2 PACKING

5.2.1 LEVEL C - A SUITABLE NUMBER OF SWIVEL ASSEMBLIES PACKAGED AS DESCRIBED IN 5.1.1 SHALL BE PACKED IN A MANNER WHICH WILL PROVIDE PROTECTION FROM DAMAGE DURING DIRECT SHIPMENT FROM THE CONTRACTOR OR SUPPLY SOURCE TO THE FIRST RECEIVING ACTIVITY FOR IMMEDIATE USE. THE GROSS WEIGHT OF THE PACKED ITEMS SHALL NOT EXCEED 60 POUNDS.

5.3 MARKING - IN ADDITION TO ANY SPECIAL MARKING REQUIRED BY THE CONTRACT OR ORDER, CONTAINER MARKING IAW MIL-STD-129 IS REQUIRED WHEN THE SWIVEL ASSEMBLY ASSEMBLY IS TO BE STOCKPILED BY A GOVERNMENT AGENCY AS A UNIQUE PART. OTHERWISE, NO SPECIAL MARKING IS REQUIRED.

6. NOTES

6.1 ORDERING DATA: PROCUREMENTS SHOULD SPECIFY THE FOLLOWING:

- A. TITLE, NUMBER AND REVISION LETTER OF THIS DRAWING.
- B. EXCEPTIONS TO THIS DRAWING, APPLICABLE DRAWINGS OR OTHER REFERENCED DOCUMENTS.
- C. DESIRED COMPLETENESS AND AVAILABILITY OF INSPECTION RECORDS, INCLUDING DATA CARDS. USE A DOD FORN 1423, WITH APPROPRIATE DATA ITEM DESCRIPTION (FORM DD 1664).

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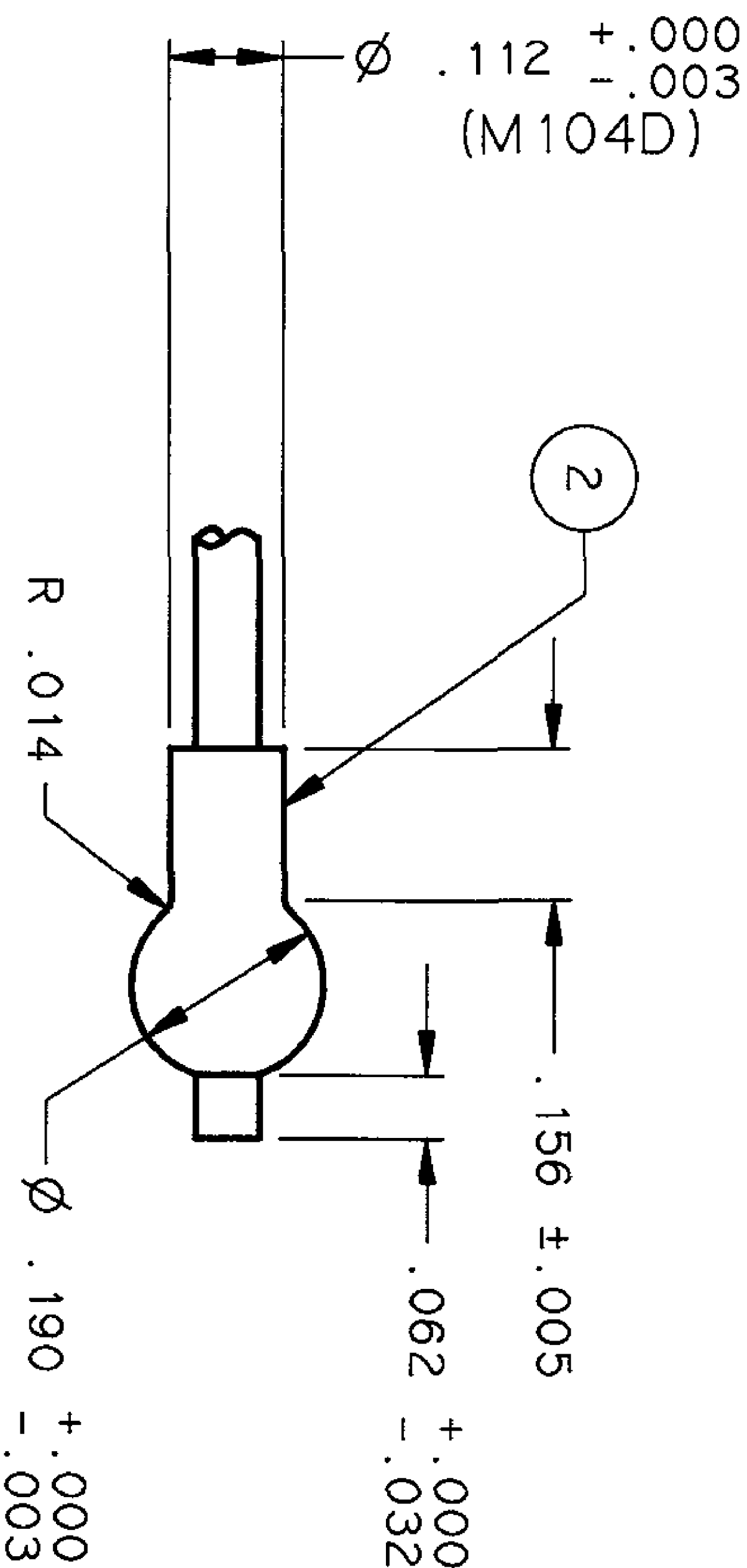
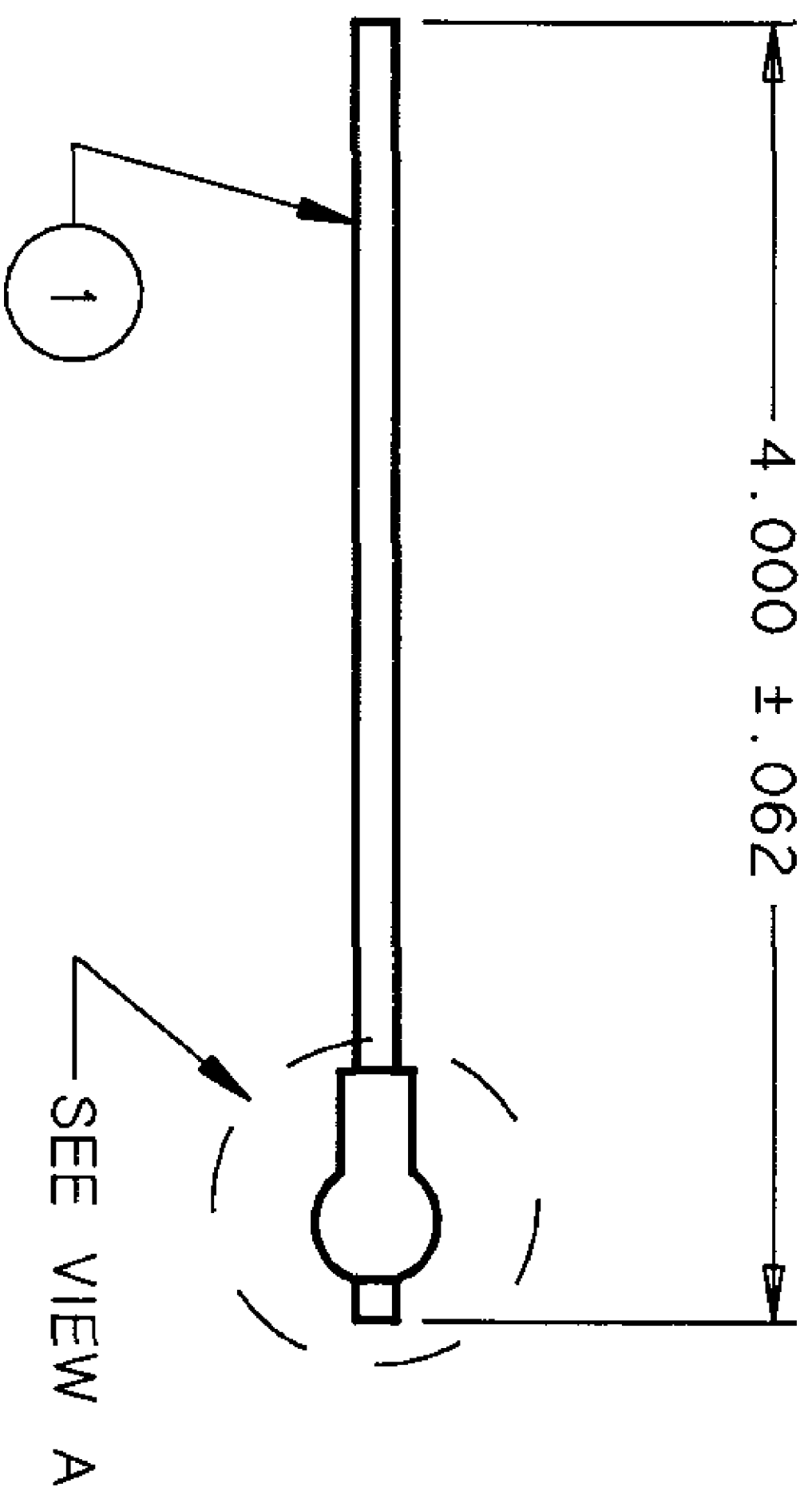
- D. TYPE AND DEGREE OF CONTRACTOR QUALITY ASSURANCE SYSTEM (MIL-1-45208 OR MIL-Q-9858).
- E. QUANTITY OF SWIVEL ASSEMBLIES REQUIRED.
- F. WHETHER OR NOT THE TEST SAMPLES REQUIRED ARE INCLUDED IN THE NUMBER OF SWIVEL ASSEMBLIES ORDERED.
- G. TESTING ACTIVITIES DESIGNATED BY THE PROCURING AGENCY TO PERFORM TESTS (SEE 4.3).
- H. DISPOSITION OF TEST SAMPLES.
- I. LEVEL OF PRESERVATION AND PACKAGING REQUIRED (SEE SECTION 5).
- J. MARKING REQUIREMENTS.
- 6.2 FIRST ARTICLE: WHEN A FIRST ARTICLE IS REQUIRED, IT, SHALL BE TESTED AND APPROVED UNDER THE APPROPRIATE PROVISIONS OF 7-104.55 OF THE ARMED SERVICES PROCUREMENT REGULATION. THE FIRST ARTICLE PROVISIONS ARE APPLICABLE AFTER AWARD OF A CONTRACT AND PRIOR TO INITIATING PRODUCTION. GENERALLY, THE PROVISIONS ARE APPLICABLE TO ONE CONTRACT; HOWEVER, THEY MAY BE WAIVED WHEN THE CONTRACTOR IS IN CONTINUOUS PRODUCTION FROM A PRIOR CONTRACT. FIRST ARTICLE SAMPLES SHALL BE REQUIRED OF CONTRACTORS.
- A. WHO HAVE NOT PREVIOUSLY PASSED FIRST ARTICLE INSPECTIONS.
- B. WHO HAVE HAD A LAPSE IN PRODUCTION FOR A PERIOD IN EXCESS OF 90 DAYS.
- C. WHO HAVE HAD A CHANGE IN MANUFACTURING PROCESS MATERIAL USE, DRAWING, SPECIFICATION OR SOURCE OF SUPPLY SO AS TO SIGNIFICANTLY AFFECT PRODUCT UNIFORMITY AS DETERMINED BY THE GOVERNMENT. THE CONTRACTING OFFICER SHOULD INCLUDE SPECIFIC INSTRUCTIONS IN ALL PROCUREMENT INSTRUMENTS, REGARDING ARRANGEMENTS FOR EXAMINATIONS, TEST AND APPROVAL OF THE FIRST ARTICLE.

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TABLE 1.
FIRST ARTICLE AND QUALITY CONFORMANCE INSPECTION

CC NUMBER	ATTRIBUTE	REQUIREMENT LOCATION	TEST METHOD OR PARA.	INSPECTION PLAN
(M101)	MINIMUM LOAD	3.2.3.1	4.5.1	4.5.1
(M102)	MAXIMUM LOAD	3.2.3.2	4.5.1	4.5.1
(M103)	SEPARATION MODE	3.2.4	4.5.1	4.5.1
(M104D)	MAX. DIA BALL SHANK	FIGURE 1	CERTIFICATE OF CONFORMANCE	EACH LOT
(M105D)	BALL END MATERIAL	FIGURE 1 FIND NO. 2	CERTIFICATE OF CONFORMANCE	EACH LOT
(M106D)	WIRE ROPE	FIGURE 1 FIND NO. 1	CERTIFICATE OF CONFORMANCE	EACH LOT
(M107D)	STEEL WIRE	FIGURE 2.	CERTIFICATE OF CONFORMANCE	EACH LOT
(M108D)	PLATING	FIGURE 2.	CERTIFICATE OF CONFORMANCE	EACH LOT
(M109D)	SWAGING SLEEVE	FIGURE 3. FIND NO. 3	CERTIFICATE OF CONFORMANCE	EACH LOT
(M110D)	INSULATION SLEEVING	FIGURE 3. FIND NO. 4	CERTIFICATE OF CONFORMANCE	EACH LOT

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VIEW A
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(M105D)	2	1	MS20664C2	BALL END, SWAGING. SINGLE SHANK
(M106D)	1	AR	MIL-C-18375	CABLE, STEEL, PREFORMED, .062 DIA, 7X7 CONSTRUCTION
FIND NO.		QTY REQD	PART OR IDENTIFYING NO.	NOMENCLATURE OR DESCRIPTION
PARTS LIST				

FIGURE 1

SWAGED BALL ASSEMBLY

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NOTES:

- (M107D) 1. MATERIAL: WIRE, STEEL, CARBON, .080 $\pm$.002 DIA PER QQ-W-428, TYPE II, CLASS 2.
- (M108D) 2. APPLY COATING, ALUMINUM, ION VAPOR DEPOSITED PER MIL-R-83488 TYPE II CLASS 2
3. SIDES OF LOOP MUST LIE BETWEEN TWO PARALLEL PLANES 0.110 MAXIMUM APART OVER THE DISTANCE SHOWN.
4. MINIMUM EXTENT OF .787 DIMENSION.

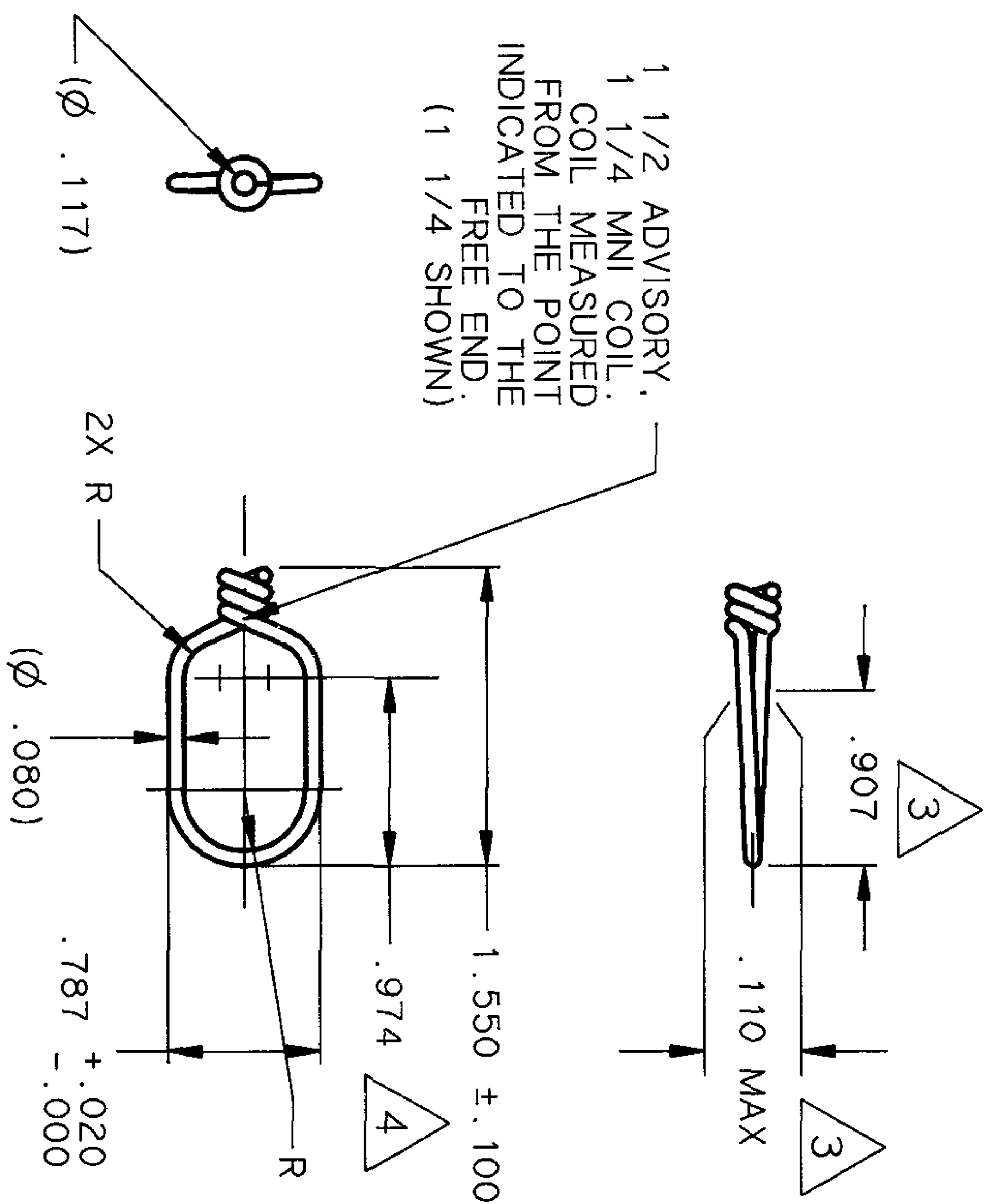


FIGURE 2

LINK

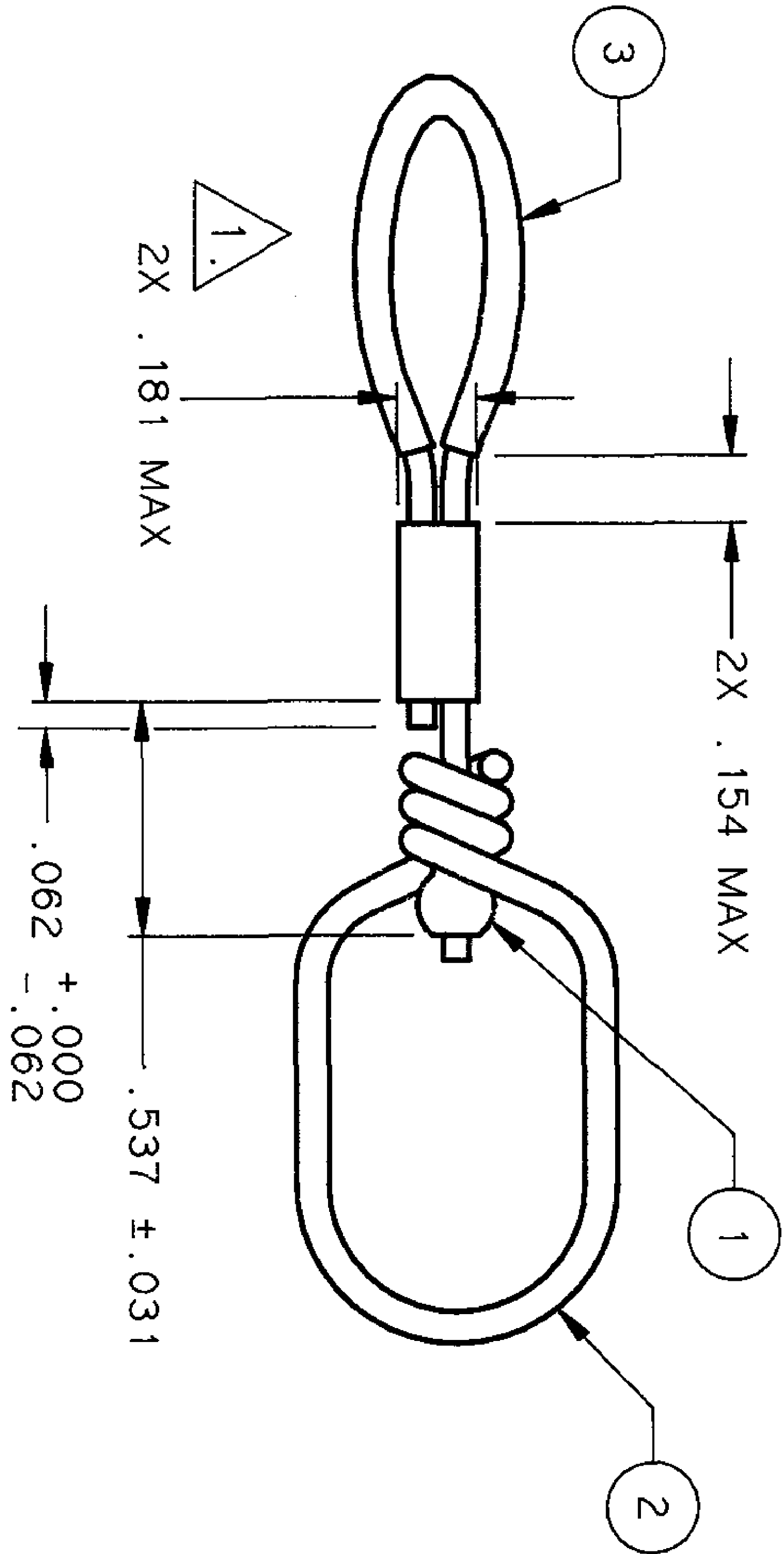
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SCALE: 1/1

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NOTES:

1. .181 ±.003 APPLIES FOR .375 MINIMUM
LENGTH AFTER SWAGING.



(M110D)		4	AR	M23053/5-104-0 PER MIL-1-23053	INSULATION SLEEVING
(M109D)		3	1	MS 51844-22	SLEEVE, SWAGING
		2	1	FIGURE 2	LINK
		1	1	FIGURE 1	SWAGED BALL ASSEMBLY
		FIND NO.	QTY	PART OR IDENTIFYING NO.	NOMENCLATURE OR DESCRIPTION
PARTS LIST					

FIGURE 3
SWIVEL ASSEMBLY

SWIVEL ASSEMBLY

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SCALE: 2/1		SHEET	12